Parametric Equations

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Derivatives of parametric equations

For $x(t) = t^2$, $y(t) = t^3$, find $\frac{dy}{dx}$ in terms of t.

$\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{3t^2}{2t} = \frac{3t}{2}$.

2 [Foundational] • Target: Derivatives of parametric equations

For $x(t) = \cos t$, $y(t) = \sin t$, find $\frac{dy}{dx}$ and evaluate it at $t = \frac{\pi}{4}$.

$\frac{dy}{dx} = \frac{\cos t}{-\sin t} = -\cot t$. At $t = \frac{\pi}{4}$: $-\cot\left(\frac{\pi}{4}\right) = -1$.

3 [Foundational] • Target: Tangent lines to parametric curves

Find the equation of the tangent line to the curve $x(t) = t^2 - 1$, $y(t) = t^3 - t$ at $t = 2$.

At $t = 2$: $x = 3$, $y = 6$. $\frac{dx}{dt} = 2t = 4$; $\frac{dy}{dt} = 3t^2 - 1 = 11$. $\frac{dy}{dx} = \frac{11}{4}$. Tangent: $y - 6 = \frac{11}{4}(x - 3)$.

4 [Foundational] • Target: Arc length of parametric curves

State the arc length formula for a parametric curve $x(t)$, $y(t)$ on $[a, b]$.

$L = \int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}\, dt$.

5 [Foundational] • Target: Arc length of parametric curves

Find the arc length of $x(t) = 3t$, $y(t) = 4t$ for $t \in [0, 2]$.

$\frac{dx}{dt} = 3$, $\frac{dy}{dt} = 4$. $L = \int_0^2 \sqrt{9 + 16}\, dt = \int_0^2 5\, dt = 10$.

6 [Foundational] • Target: Arc length of parametric curves

Set up (but do not evaluate) the arc length integral for $x(t) = t^2$, $y(t) = t^3$ on $[0, 1]$.

$\frac{dx}{dt} = 2t$, $\frac{dy}{dt} = 3t^2$. $L = \int_0^1 \sqrt{(2t)^2 + (3t^2)^2}\, dt = \int_0^1 \sqrt{4t^2 + 9t^4}\, dt$.

7 [Foundational] • Target: Speed as a vector quantity

A particle moves with position $x(t) = t^2$, $y(t) = 2t$. Find its speed at $t = 1$.

$\frac{dx}{dt} = 2t$, $\frac{dy}{dt} = 2$. At $t = 1$: $\frac{dx}{dt} = 2$, $\frac{dy}{dt} = 2$. Speed $= \sqrt{2^2 + 2^2} = \sqrt{8} = 2\sqrt{2}$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

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8 [Intermediate] • Target: Speed as a vector quantity

A particle moves with position $x(t) = 3\cos t$, $y(t) = 3\sin t$. Find its speed at any time t, and interpret the result.

$\frac{dx}{dt} = -3\sin t$, $\frac{dy}{dt} = 3\cos t$. Speed $= \sqrt{9\sin^2 t + 9\cos^2 t} = \sqrt{9} = 3$ (constant). The particle traces a circle of radius 3 at constant speed.

9 [Intermediate] • Target: Eliminating the parameter

Eliminate the parameter from $x(t) = t$, $y(t) = t^2 - 1$ to find the Cartesian equation, and sketch it.

Since $x = t$, substitute directly: $y = x^2 - 1$, an upward parabola shifted down 1 unit.

10 [Intermediate] • Target: Eliminating the parameter

Eliminate the parameter from $x(t) = 2t + 1$, $y(t) = 4t - 3$ to find the Cartesian equation.

From $x = 2t + 1$: $t = \frac{x-1}{2}$. Substitute: $y = 4\left(\frac{x-1}{2}\right) - 3 = 2(x-1) - 3 = 2x - 5$. A line with slope 2.

11 [Intermediate] • Target: Eliminating the parameter

Explain why eliminating the parameter from $x(t) = \cos t$, $y(t) = \sin t$ using the identity $\sin^2 t + \cos^2 t = 1$ gives a relation that is NOT a function $y = f(x)$.

Substituting gives $x^2 + y^2 = 1$, the equation of a unit circle. For most x-values in $(-1, 1)$, there are two corresponding y-values ($\pm\sqrt{1 - x^2}$), so this relation fails the vertical line test and is not a function of x.

12 [Intermediate] • Target: Concavity of parametric curves

For $x(t) = t^2$, $y(t) = t^3$ (with $\frac{d^2y}{dx^2} = \frac{3}{4t}$ found earlier), determine the concavity of the curve for $t > 0$ and for $t < 0$.

For $t > 0$: $\frac{d^2y}{dx^2} = \frac{3}{4t} > 0$, so the curve is concave up. For $t < 0$: $\frac{3}{4t} < 0$, so the curve is concave down.

13 [Intermediate] • Target: Horizontal and vertical tangents

For $x(t) = t^2 - 4$, $y(t) = t^3 - 3t$, find the value(s) of t where the tangent line is horizontal (where $\frac{dy}{dt} = 0$, provided $\frac{dx}{dt} \neq 0$).

$\frac{dy}{dt} = 3t^2 - 3 = 0 \Rightarrow t = \pm 1$. Check $\frac{dx}{dt} = 2t$: at $t = \pm 1$, $\frac{dx}{dt} = \pm 2 \neq 0$, so both give valid horizontal tangents.

14 [Intermediate] • Target: Horizontal and vertical tangents

For the same curve, find the value(s) of t where the tangent line is vertical (where $\frac{dx}{dt} = 0$, provided $\frac{dy}{dt} \neq 0$).

$\frac{dx}{dt} = 2t = 0 \Rightarrow t = 0$. Check $\frac{dy}{dt} = 3(0)^2 - 3 = -3 \neq 0$, so $t = 0$ gives a valid vertical tangent, at the point $x(0) = -4$, $y(0) = 0$.

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Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: Motion along a parametric path (particle interpretation)** • **Cross-Lesson: 5.02 Ratio, Root, and Alternating Series Tests**

A particle's position is $x(t) = t^2$, $y(t) = t^3 - 3t$ for $t \in [-2, 2]$. Find all times when the particle is moving straight up or down (vertical velocity component only, i.e. $\frac{dx}{dt} = 0$).

$\frac{dx}{dt} = 2t = 0 \Rightarrow t = 0$. At $t = 0$: position is $(0, 0)$, and the particle is momentarily moving purely vertically (since horizontal velocity is 0 there).

16 **[Challenging]** • **Target: Motion along a parametric path (particle interpretation)** • **Cross-Lesson: 6.01 Power Series and Interval of Convergence**

For the particle above, find the speed at $t = 2$.

$\frac{dx}{dt} = 2t = 4$; $\frac{dy}{dt} = 3t^2 - 3 = 9$. Speed $= \sqrt{4^2 + 9^2} = \sqrt{16 + 81} = \sqrt{97} \approx 9.85$.

17 **[Challenging]** • **Target: Arc length of a full circle (verifying the circumference formula)** • **Cross-Lesson: 6.02 Taylor and Maclaurin Series**

Use the parametric arc length formula to verify that the circumference of a circle $x(t) = r\cos t$, $y(t) = r\sin t$ for $t \in [0, 2\pi]$ is $2\pi r$.

$\frac{dx}{dt} = -r\sin t$, $\frac{dy}{dt} = r\cos t$. $L = \int_0^{2\pi} \sqrt{r^2\sin^2 t + r^2\cos^2 t}\, dt = \int_0^{2\pi} r\, dt = r(2\pi) = 2\pi r$, matching the known circumference formula.

18 **[Challenging]** • **Target: Arc length of a full circle (verifying the circumference formula)** • **Cross-Lesson: 1.01 Integration by Parts**

Find the arc length of the parametric curve $x(t) = e^t\cos t$, $y(t) = e^t\sin t$ is a spiral; set up (but do not evaluate) its arc length integral on $[0, \pi]$.

$\frac{dx}{dt} = e^t\cos t - e^t\sin t$, $\frac{dy}{dt} = e^t\sin t + e^t\cos t$. After simplification, $\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 = 2e^{2t}$. So $L = \int_0^{\pi} \sqrt{2e^{2t}}\, dt = \int_0^{\pi} e^t\sqrt{2}\, dt$.

19 **[Challenging]** • **Target: Parametric equations from real motion** • **Cross-Lesson: 1.02 Partial Fractions and Improper Integrals**

A Ferris wheel of radius 10 m completes one revolution every 40 s, centered 12 m above the ground. Write parametric equations for a rider's height above the ground as a function of time t (seconds), assuming the rider starts at the bottom.

Angular speed $= \frac{2\pi}{40} = \frac{\pi}{20}$ rad/s. Starting at the bottom means height $= 12 - 10\cos\left(\frac{\pi}{20}t\right)$ (using $x(t) = t$ for time, $y(t) = 12 - 10\cos\left(\frac{\pi}{20}t\right)$ for height).

20 **[Challenging]** • **Target: Parametric equations from real motion** • **Cross-Lesson: 2.01 Euler's Method and Logistic Growth**

For the Ferris wheel model above, find the rider's vertical velocity $\frac{dy}{dt}$ at $t = 10$ s (a quarter revolution).

$\frac{dy}{dt} = 10 \cdot \frac{\pi}{20}\sin\left(\frac{\pi}{20}t\right) = \frac{\pi}{2}\sin\left(\frac{\pi}{20}t\right)$. At $t = 10$: $\frac{\pi}{2}\sin\left(\frac{\pi}{2}\right) = \frac{\pi}{2} \approx 1.57$ m/s.

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Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Derivatives of parametric equations

Eliminate the parameter from $x(t) = -2t - 3$, $y(t) = 7t - 9$ to find the Cartesian equation, then state its slope.

a. $-\frac{7}{2}$ ✓

b. $-\frac{5}{2}$

c. $-\frac{2}{7}$

d. $\frac{7}{2}$

From $x = -2t - 3$: $t = \frac{x+3}{-2}$. Substituting into $y = 7t - 9$ gives a line of slope $\frac{7}{-2} = -\frac{7}{2}$.

22 [Foundational] • Target: Second derivatives of parametric equations

For $x(t) = t^2 + 2t$, $y(t) = t^3 + 6t^2$, find the larger value of t where the tangent line is horizontal.

a. -4

b. 0 ✓

c. -3

d. 3

$\frac{dy}{dt} = 3t^2 + 12t = 3(t - (-4))(t - (0)) = 0$ gives $t = -4$ and $t = 0$; the larger is 0.

23 [Foundational] • Target: Tangent lines to parametric curves

A particle's motion is described by the parametric equations $x(t) = e^{2t} - 2t$ and $y(t) = 4e^t$ for $t \geq 0$. Find the value of $\frac{dy}{dx}$ at the point where the tangent line to the curve is horizontal.

a. **0** ✓

b. 1

c. 2

d. 4

To find the slope of the tangent line, we compute $\frac{dy}{dx} = \frac{dy/dt}{dx/dt}$. \\ First, we find the derivatives with respect to t: \\ $\frac{dx}{dt} = \frac{d}{dt}(e^{2t} - 2t) = 2e^{2t} - 2$ \\ $\frac{dy}{dt} = \frac{d}{dt}(4e^t) = 4e^t$ \\ So, $\frac{dy}{dx} = \frac{4e^t}{2e^{2t} - 2}$. \\ A horizontal tangent line occurs when $\frac{dy}{dt} = 0$ and $\frac{dx}{dt} eq 0$. \\ Setting $\frac{dy}{dt} = 0$, we get $4e^t = 0$. This equation has no solution for real t since $e^t > 0$ for all real t. \\.

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24 [Foundational] • Target: Arc length of parametric curves

A particle moves along $x(t) = 2t^2,\ y(t) = 1t^3$. Find its speed at $t = 2$.

a. $1 + 4\sqrt{13}$

b. $2\sqrt{13}$

c. $4\sqrt{13}$ ✓

d. 20

$x' = 4t,\ y' = 3t^2$. Speed $= \sqrt{(x')^2 + (y')^2}\,|_{t=2} = 4\sqrt{13}$.

25 [Foundational] • Target: Speed as a vector quantity

A particle moves along parametric path $x(t) = 3t^2$, $y(t) = 5t$. What is the velocity vector $\mathbf{v}(t) = \langle x'(t), y'(t) \rangle$?

a. $\langle 6t, 5 \rangle$ ✓

b. $\langle 3t, 5 \rangle$

c. $\langle 6t, 0 \rangle$

d. $\langle 3t^2, 5t \rangle$

Differentiating component-wise gives $x'(t) = 6t$ and $y'(t) = 5$.

26 [Foundational] • Target: Eliminating the parameter

Find the arc length of $x = 3t^2,\ y = \frac{6}{3}t^3$ from $t = 0$ to $t = 2$.

a. $-1 + 5\sqrt{5}$

b. $-4 + 20\sqrt{5}$

c. $-2 + 10\sqrt{5}$ ✓

d. $-1 + 10\sqrt{5}$

$x' = 6t,\ y' = 6t^2$. $L = \int_0^2 \sqrt{(x')^2 + (y')^2}\,dt = -2 + 10\sqrt{5}$.

27 [Foundational] • Target: Concavity of parametric curves

At a certain instant, a particle has velocity components $x'(t) = 3$ and $y'(t) = 4$. Find its speed.

a. $2\sqrt{3}$

b. **5** ✓

c. 7

d. 25

Speed $= \sqrt{(x')^2 + (y')^2} = \sqrt{3^2 + 4^2} = 5$.

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28 [Foundational] • Target: Horizontal and vertical tangents

Arc length of parametric curve: $L = \int_a^b$

a. $\sqrt{(dx/dt) + (dy/dt)}\, dt$

b. $\sqrt{(dx/dt)^2 + (dy/dt)^2}\, dt$ ✓

c. $(dx/dt)^2 + (dy/dt)^2\, dt$

d. $|dy/dx|, dt$

$L = \int_a^b \sqrt{\dot{x}^2 + \dot{y}^2}\, dt$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Motion along a parametric path (particle interpretation)

A particle's position at time t is given by the vector-valued function $\mathbf{r}(t) = \langle 3t^2 + 5,\ -2t + 1\rangle$. Find the velocity vector $\mathbf{v}(t)$ at time t.

a. $\langle 3t^2 + 5,\ -2t + 1\rangle$

b. $\langle 6t,\ -2\rangle$ ✓

c. $\langle 3t,\ -2\rangle$

d. $\langle 6t^2,\ -2t\rangle$

The position vector is given by $\mathbf{r}(t) = \langle x(t), y(t)\rangle = \langle 3t^2 + 5,\ -2t + 1\rangle$. To find the velocity vector, we differentiate each component with respect to t. The derivative of $x(t) = 3t^2 + 5$ is $x'(t) = 6t$. The derivative of $y(t) = -2t + 1$ is $y'(t) = -2$. Therefore, the velocity vector is $\mathbf{v}(t) = \langle x'(t), y'(t)\rangle = \langle 6t,\ -2\rangle$.

30 [Intermediate] • Target: Arc length of a full circle (verifying the circumference formula)

A particle's position is described by the parametric equations $x(t) = 3t + 5$ and $y(t) = t^2 - 2$. What is the velocity vector $\mathbf{v}(t) = \langle x'(t), y'(t)\rangle$ at time t?

a. $\langle 3, 2t\rangle$ ✓

b. $\langle 3, t^2 - 2\rangle$

c. $\langle 3t + 5, 2t\rangle$

d. $\langle 3, 2\rangle$

To find the velocity vector, we need to find the derivatives of the parametric equations with respect to t. The derivative of $x(t) = 3t + 5$ is $x'(t) = \frac{d}{dt}(3t + 5) = 3$. The derivative of $y(t) = t^2 - 2$ is $y'(t) = \frac{d}{dt}(t^2 - 2) = 2t$. Therefore, the velocity vector is $\mathbf{v}(t) = \langle x'(t), y'(t)\rangle = \langle 3, 2t\rangle$.

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31 [Intermediate] • Target: Parametric equations from real motion

Consider a particle whose position is described by the parametric equations $x(t) = 3t + 5$ and $y(t) = 2t^2 - 1$ for $t \geq 0$. What is the velocity vector $\mathbf{v}(t) = \langle x'(t), y'(t) \rangle$?

a. $\langle 3, 4t \rangle$ ✓

b. $\langle 3t, 4t \rangle$

c. $\langle 3, 2t \rangle$

d. $\langle 3, 4t - 1 \rangle$

The velocity vector is found by differentiating the parametric equations with respect to t. The derivative of $x(t) = 3t + 5$ is $x'(t) = 3$. The derivative of $y(t) = 2t^2 - 1$ is $y'(t) = 4t$. Therefore, the velocity vector is $\mathbf{v}(t) = \langle x'(t), y'(t) \rangle = \langle 3, 4t \rangle$.

32 [Intermediate] • Target: Derivatives of parametric equations

A particle has velocity $\mathbf{r}'(t) = \langle 1,\ 3t \rangle$ and position $\mathbf{r}(0) = \langle 2,\ 1 \rangle$. Find $\mathbf{r}(1)$.

a. $\langle 3,\ 5/2 \rangle$ ✓

b. $\langle 2,\ 1 \rangle$

c. $\langle 5/2,\ 3 \rangle$

d. $\langle 4,\ 5/2 \rangle$

$x(t) = 2 + \int_0^t 1\, ds = t + 2$, $y(t) = 1 + \int_0^t 3s\, ds = \frac{3t^2}{2} + 1$. At $t = 1$: $\langle 3, 5/2 \rangle$.

33 [Intermediate] • Target: Second derivatives of parametric equations

A particle moves with $x(t) = 1\cos t,\ y(t) = 1\sin t$ from $t = 0$ to $t = 2\pi$. Find the total distance travelled.

a. 4π

b. π

c. 2π ✓

d. 1

Speed $= \sqrt{x'^2 + y'^2} = 1$ (constant). Distance $= 1 \times (2\pi - (0)) = 2\pi$.

34 [Intermediate] • Target: Tangent lines to parametric curves

A particle moves with $x(t) = 3t^3,\ y(t) = 2t^2$. Find the acceleration vector at $t = 1$.

a. $\langle 18,\ 5 \rangle$

b. $\langle 19,\ 4 \rangle$

c. $\langle 4,\ 18 \rangle$

d. $\langle 18,\ 4 \rangle$ ✓

$x''(t) = 18t,\ y''(t) = 4$. At $t = 1$: $\langle 18, 4 \rangle$.

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35 [Intermediate] • Target: Arc length of parametric curves

A particle moves with $x(t) = 1t^2,\ y(t) = 2t$. Find the velocity vector at $t = 1$.

a. $\langle 4,\ 2\rangle$

b. $\langle 3,\ 2\rangle$

c. $\langle 1,\ 2\rangle$

d. $\langle 2,\ 2\rangle$ ✓

$x'(t) = 2t,\ y'(t) = 2$. At $t = 1$: $\langle 2, 2\rangle$.

36 [Intermediate] • Target: Speed as a vector quantity

A particle has velocity vector $\mathbf{r}'(t) = \langle 9,\ 12\rangle$ at a given instant. Compute $r'(t)$.

a. 21

b. **15** ✓

c. 225

d. $3\sqrt{7}$

$'(t) = 9^2 + 12^2 = 15$.

37 [Intermediate] • Target: Eliminating the parameter

Find the slope of the tangent line $\frac{dy}{dx}$ to the parametric curve $x(t) = t^2 + 1,\ y(t) = t^3 - 3t$ at $t = 2$.

a. $\frac{9}{4}$ ✓

b. $\frac{11}{4}$

c. $\frac{4}{9}$

d. $\frac{8}{4}$

$\frac{dy}{dx} = \frac{y'(t)}{x'(t)} = \frac{3t^2 - 3}{2t}$. At $t = 2$, $\frac{3(4) - 3}{2(2)} = \frac{9}{4}$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Concavity of parametric curves • Cross-Lesson: 1.01 Integration by Parts

A curve is given by $x(t) = t^2,\ y(t) = t^3 - 3t$. Find $\frac{dy}{dx}$ at $t = 1$.

a. 3

b. **0** ✓

c. 1

d. 2

$\frac{dx}{dt} = 2t,\ \frac{dy}{dt} = 3t^2 - 3$. At $t = 1$: $\frac{dy}{dx} = \frac{0}{2} = 0$.

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39 [Challenging] • Target: Horizontal and vertical tangents • Cross-Lesson: 1.02 Partial Fractions and Improper Integrals

The cross product of two parallel vectors is:

a. A unit vector

b. **The zero vector ✓**

c. Their dot product

d. Undefined

Parallel vectors give $\vec{0}$ cross product.

40 [Challenging] • Target: Motion along a parametric path (particle interpretation) • Cross-Lesson: 2.01 Euler's Method and Logistic Growth

Find the arc length of the spiral $r = 3\theta$ from $\theta = 0$ to $\theta = 2\pi$.

a. $3\operatorname{asinh}(2\pi) + 6\pi\sqrt{1+4\pi^2}$

b. $\frac{3\operatorname{asinh}(2\pi)}{2} + 3\pi\sqrt{1+4\pi^2}$ ✓

c. 6π

d. $\frac{3\operatorname{asinh}(2\pi)}{4} + \frac{3\pi\sqrt{1+4\pi^2}}{2}$

Arc length $= \int_0^{2\pi} \sqrt{r^2 + (dr/d\theta)^2}\, d\theta = \int_0^{2\pi} \sqrt{3^2\theta^2 + 3^2}\, d\theta = \frac{3\operatorname{asinh}(2\pi)}{2} + 3\pi\sqrt{1+4\pi^2}$.

41 [Challenging] • Target: Arc length of a full circle (verifying the circumference formula) • Cross-Lesson: 3.02 Vector-Valued Functions

Find the total area enclosed by the polar cardioid $r = 2(1+\cos\theta)$ for $0 \le \theta \le 2\pi$.

a. 4π

b. $\frac{12\pi}{2}$ ✓

c. $2 \cdot 4\pi$

d. $\frac{12\pi}{4}$

Area $A = \frac{1}{2}\int_0^{2\pi} r^2 d\theta = \frac{2^2}{2}\int_0^{2\pi} (1 + 2\cos\theta + \cos^2\theta) d\theta = \frac{3 \cdot 2^2 \pi}{2} = \frac{12\pi}{2}$.

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42 [Challenging] • Target: Parametric equations from real motion • Cross-Lesson: 4.01 Polar Curves and Derivatives

Consider a particle whose velocity vector $\mathbf{v}(t) = \langle x'(t), y'(t) \rangle$ satisfies the differential equation $y'' - 2y' + y = 0$ and the initial conditions $x(0) = 1, y(0) = 2, x'(0) = 0, y'(0) = 1$. Furthermore, the particle's path is such that the magnitude of its acceleration vector is always equal to the magnitude of its velocity vector, i.e., $||\mathbf{a}(t)|| = ||\mathbf{v}(t)||$. Find the value of $x(1)$.

a. $e^{-1} + 1$

b. $2e^{-1} + 1$ ✓

c. $e^{-1} - 1$

d. $2 - e^{-1}$

The given differential equation for the y-component of the velocity is $y'' - 2y' + y = 0$. The characteristic equation is $r^2 - 2r + 1 = 0$, which factors as $(r - 1)^2 = 0$. Thus, we have a repeated root $r = 1$. The general solution for $y(t)$ is $y(t) = c_1 e^t + c_2 te^t$. Using the initial conditions $y(0) = 2$ and $y'(0) = 1$: $y(0) = c_1 e^0 + c_2(0)e^0 = c_1 = 2$. $y'(t) = 2e^t + 2te^t + c_2 e^t + c_2 te^t = (2 + c_2)e^t + (2 + c_2)te^t$. $y'(0) = (2 + c_2)e^0 + (2 + c_2)(0)e^0 = 2 + c_2 = 1$, so $c_2 = -1$. Thus.

43 [Challenging] • Target: Derivatives of parametric equations • Cross-Lesson: 4.02 Area in Polar Coordinates

A particle's position in the xy-plane is given by the parametric equations $x(t) = 2t + 1$ and $y(t) = t^2 - 3$, where t is time. What is the Cartesian equation of the path traced by the particle?

a. $y = \frac{1}{2}x^2 - 3$

b. $y = \left(\frac{x-1}{2}\right)^2 - 3$ ✓

c. $y = x^2 - 3$

d. $y = 2\left(\frac{x-1}{2}\right) + 1$

To find the Cartesian equation, we need to eliminate the parameter t from the given parametric equations. Given: $x(t) = 2t + 1$ $y(t) = t^2 - 3$ From the first equation, we can solve for t in terms of x: $x - 1 = 2t$ $t = \frac{x-1}{2}$ Now, substitute this expression for t into the second equation: $y = \left(\frac{x-1}{2}\right)^2 - 3$ This is the Cartesian equation of the path traced by the particle.

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44 [Challenging] • Target: Second derivatives of parametric equations • Cross-Lesson: 5.01 Sequences and Basic Series Convergence Tests

Epicycloid with n cusps (outer radius R, small circle radius $r = R/n$) has arc length:

a. $8nR$

b. $8R$ ✓

c. $4nR$

d. $2\pi nR$

Epicycloid n cusps: length $= 8R(1 + 1/n) \cdot n/(n + 1)$... actually $L = 8Rn/(n/(n + 1))$... the cardioid ($n = 1$) has length $8r = 8(R/2) = 4R$... Let me use cardioid: $r = R/2$, $L = 8(R/2) = 4R$ not 8R. For general n: $L = 8nR \cdot 1/(n + 1) \cdot (n + 1) = 8R$... The answer choice that matches the cardioid result $L = 8a$ for $r = a$ generating circle is $8R$ — let's accept that as the answer for $n = 1$ case (cardioid).

45 [Challenging] • Target: Tangent lines to parametric curves • Cross-Lesson: 5.02 Ratio, Root, and Alternating Series Tests

Arc length of $r = 2\cos\theta$ (full circle, $0 \le \theta \le \pi$):

a. π

b. 4

c. $\pi/2$

d. 2π ✓

Circle radius 1 (centered at $(1, 0)$), circumference $= 2\pi$.

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